

Modelling of Large-Scale Circulation in Atmosphere and Ocean
(MO8004)

Kelvin Waves

Lab 2

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1 Theory

(i) No equations, what is it? what are the requremenets?

(ii) Equations. B.C.

iii Write solh

2 Experiments

2.1 Coastal Kelvin Waves

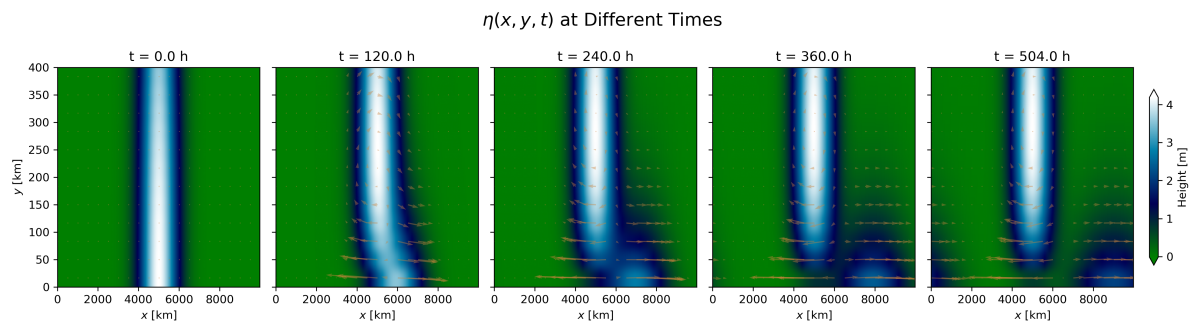
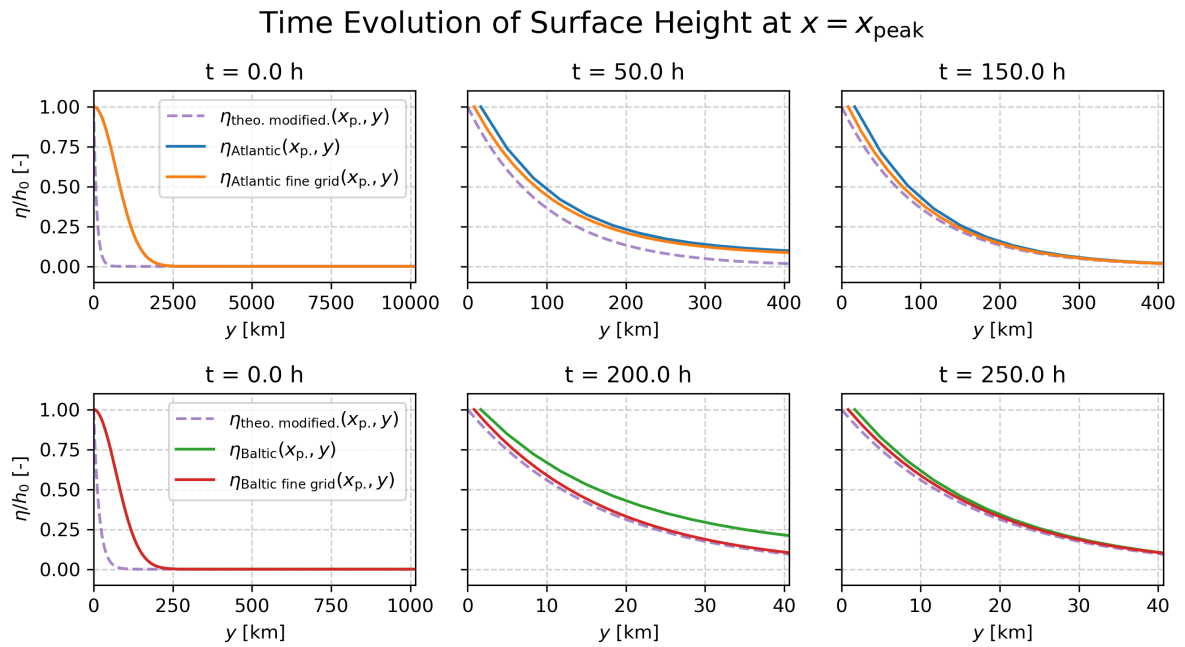
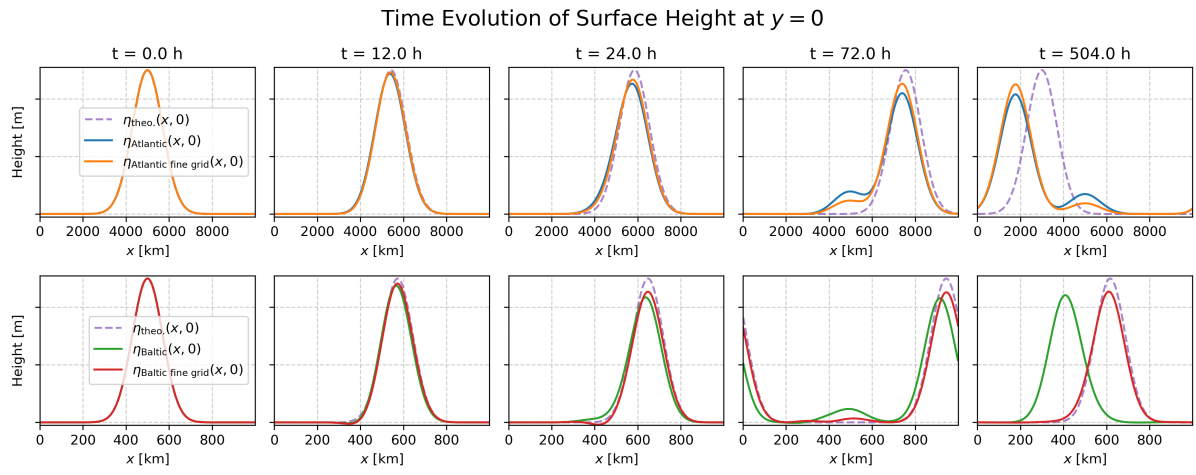


Figure 1: This figure...

Case	Time (h)	c_{num}	c_{theo}	Rel. error (%)
Atlantic	0 → 1	12.6758	9.9045	27.98
	0 → 10	10.7313	9.9045	8.35
	0 → 24	11.4026	9.9045	15.12
	0 → 168	10.5924	9.9045	6.94
	0 → 336	10.6227	9.9045	7.25
	0 → 504	10.5740	9.9045	6.76
Atlantic fine grid	0 → 1	16.1560	9.9045	63.12
	0 → 10	10.1608	9.9045	2.59
	0 → 24	10.7839	9.9045	8.88
	0 → 168	10.5634	9.9045	6.65
	0 → 336	10.5848	9.9045	6.87
	0 → 504	10.5689	9.9045	6.71
Baltic	0 → 1	2.8254	1.7155	64.70
	0 → 10	1.7792	1.7155	3.71
	0 → 24	1.8224	1.7155	6.23
	0 → 168	1.8263	1.7155	6.46
	0 → 336	1.8279	1.7155	6.55
	0 → 504	1.8303	1.7155	6.69
Baltic fine grid	0 → 1	3.1741	1.7155	85.03
	0 → 10	1.6765	1.7155	2.28
	0 → 24	1.7226	1.7155	0.41
	0 → 168	1.7192	1.7155	0.22
	0 → 336	1.7185	1.7155	0.17
	0 → 504	1.7179	1.7155	0.14

Table 1: Kelvin wave phase speed: numerical vs theoretical comparison.

Figure 2: This figure...**Figure 3:** This figure...

**Figure 4:** This figure...

2.2 Equatorial Kelvin Waves

(i)

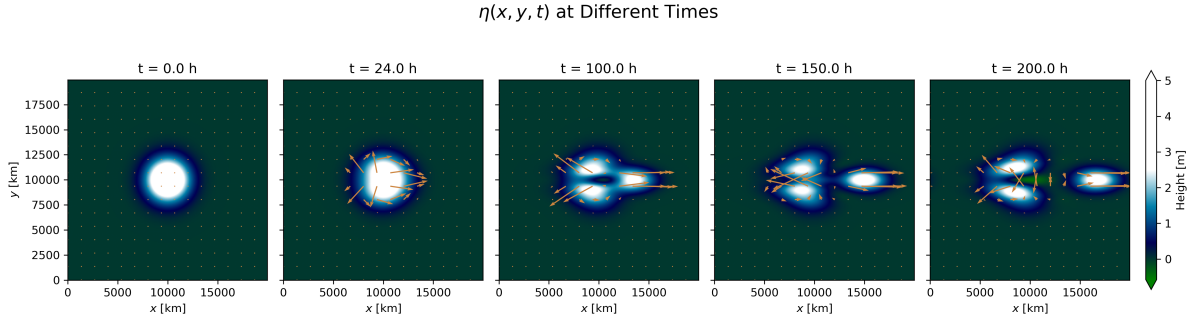


Figure 5: This figure...

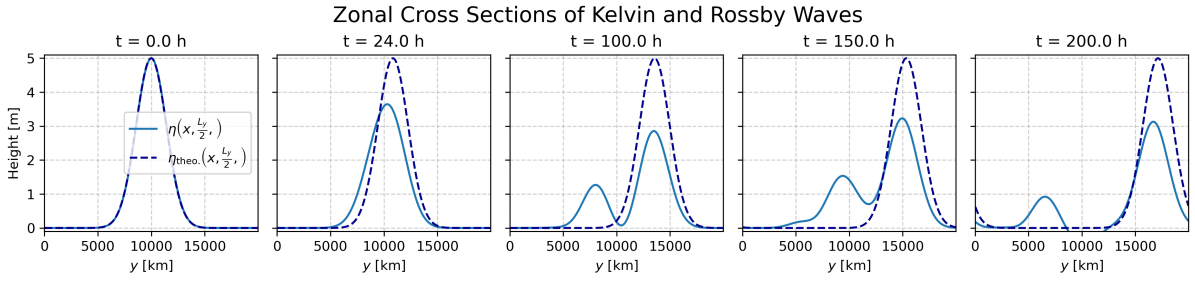


Figure 6: This figure...

Time [h]	$x_{\max}$ [km]	c_{num} [m/s]	c_{theo} [m/s]	Error [%]
0 → 24	10300.0	3.86	9.90	61.05
24 → 100	13500.0	11.70	9.90	18.09
100 → 150	14966.7	8.15	9.90	17.73
150 → 200	16700.0	9.63	9.90	2.78
0 → 200	16700.0	9.33	9.90	5.81

Table 2: Kelvin-wave phase speed diagnostics taken from Figure 6.

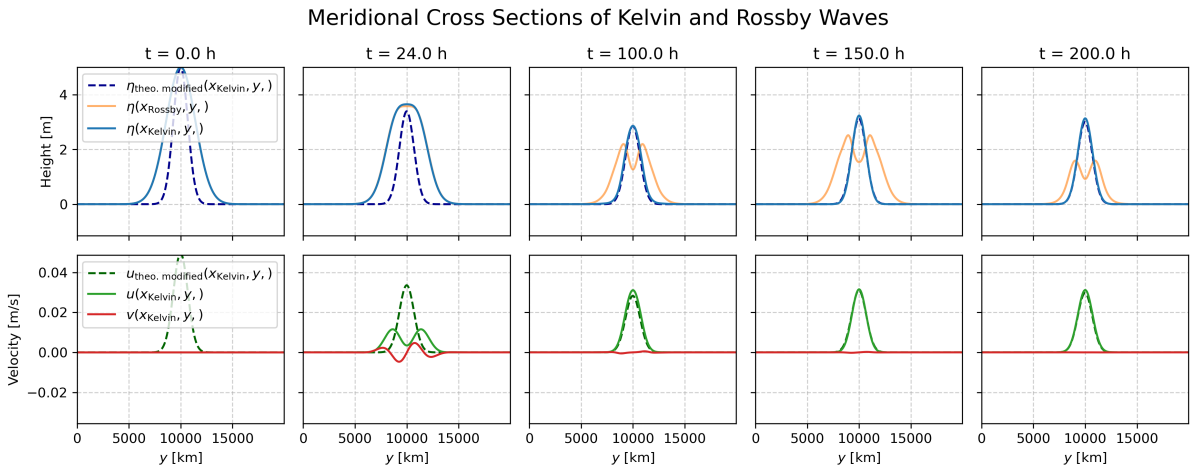


Figure 7: This figure...

A Appendix

The source code for this lab can be found on GitHub:

`https://github.com/FredrikBergelv/
Modelling-of-Large-Scale-Circulation-in-Atmosphere-and-Ocean`